

Incorporating AI into In-Situ Process Monitoring for Fault Detection

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ABSTRACT

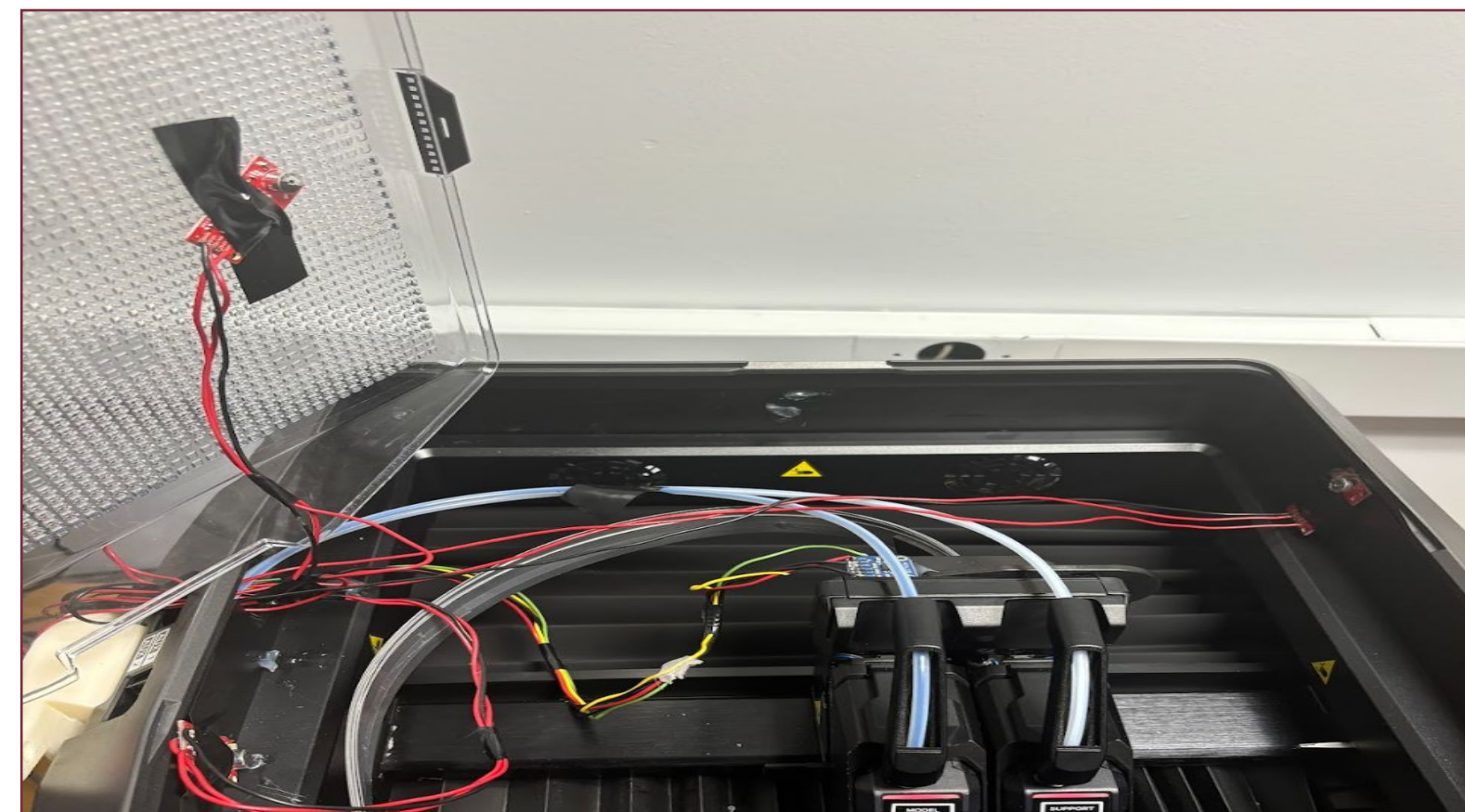
Manufacturing all over the world today have started to incorporate 3D printers into their system. 3D printers have become a necessity to modern manufacturing processes, offering efficiency. The usage of acoustic sensors in 3D printers enable for creating a dataset and processing the audio from 3D printers. Machine learning can play a huge role in increasing the efficiency of 3D printers with ML being able to learn from a dataset and classify anomalies.

OBJECTIVES

- Record a dataset of 3D print audio.
- Classify normal audio vs anomaly.
- Detect anomaly from a 3D print audio.
- Use a deep learning model such as LSTM to classify.
- Make 3D Printers more efficient.

METHODS

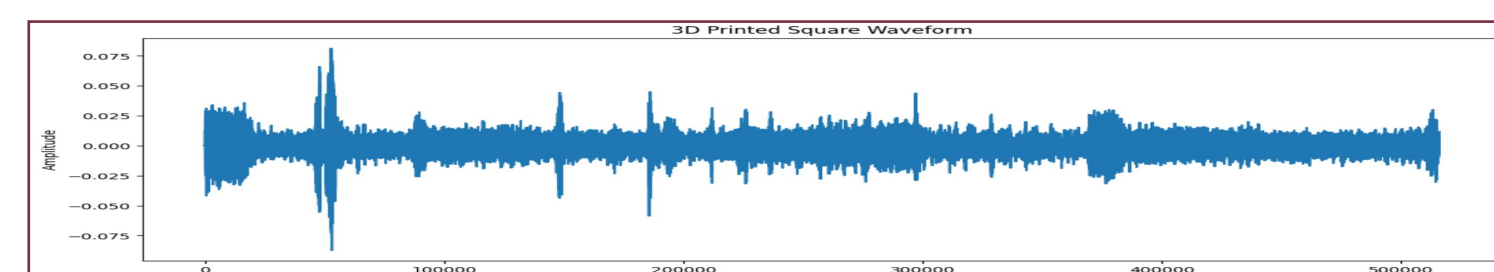
- Set up acoustic sensors inside of 3D Printer
- Put in an angular array for accurate beamforming effect.
- Record audio from 3D prints and process it with spectrograms and waveforms.
- Create a dataset to train ML model.
- ML model is used to classify.



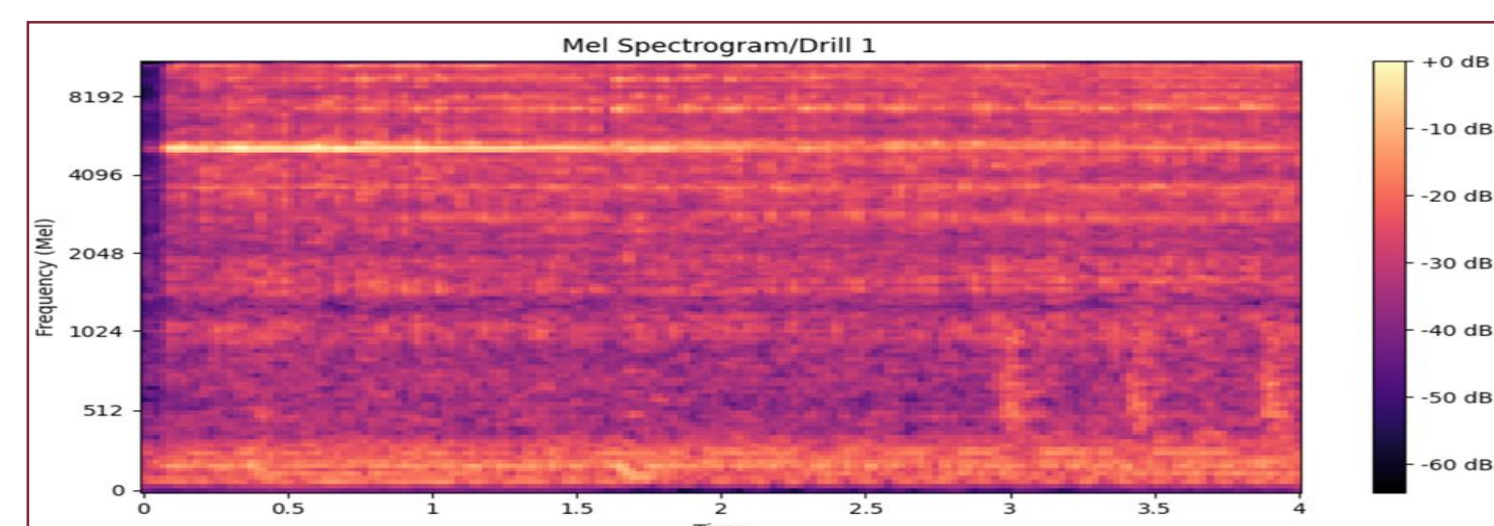
Angular microphone array inside 3D printer

PROCESSING

- Processed audio from 3D printer using google colab and Friture.
- Looked at waveforms and Mel Spectrogram
- Extract MFCC for training



Graphs of the processed data, top showing the waveform, while bottom is Mel Spectrogram



GOOGLE COLAB/ML

- Process the audio by extracting important features such as MFCC.
- Split the features into two array, independent vs dependent.
- Train the deep learning model.
- Got an .81 accuracy.

```
test_accuracy=model.evaluate(X_test,y_test,verbose=0)
print(test_accuracy[1])
```

0.812142014503479

RESULTS

Created a deep learning model able to classify an anomaly within a print.

```
filename="/content/drive/MyDrive/UrbanSound8K/UrbanSound8K/audio/fold2/100652-3-0-0.wav"
audio, sample_rate = librosa.load(filename, res_type='kaiser_fast')
mfccs_features = librosa.feature.mfcc(y=audio, sr=sample_rate, n_mfcc=40)
mfccs_scaled_features = np.mean(mfccs_features.T,axis=0)
print(mfccs_scaled_features)
mfccs_scaled_features=mfccs_scaled_features.reshape(1,-1)
print(mfccs_scaled_features)
print(mfccs_scaled_features.shape)
predicted_label = np.argmax(model.predict(mfccs_scaled_features), axis=-1)
print(predicted_label)
prediction_class = labelencoder.inverse_transform(predicted_label)
prediction_class
```

```
1/1 [=====] - 0s 43ms/step
[3]
array(['anomaly_print'], dtype='<U16')

```

Result of the model and the classification of faulty print

CONCLUSION

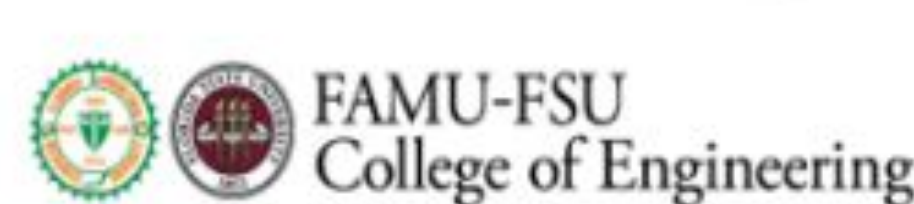
Incorporating AI/ML into 3D printing allowed for accurate fault detection by processing and training the audio data recorded from 3D prints. Using the acoustic sensors put inside the 3D printer, we were able to record data for processing, and once processed, trained the model. The model proved to have a 81% value accuracy, and was able to detect faults within the audio of a 3D Print, and this could also be used for other manufacturing purposes than just 3D printers.

REFERENCES

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ACKNOWLEDGEMENTS

The team would like to acknowledge the collaborative efforts of Asterix, KCNSC, FAMU-FSU College of Engineering, Dr. Bernadin, Fasih Waheed and Nancy Joseph for providing this opportunity.



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